```
======================================================================
EXPERIMENT 6 — ORIGINAL EXP 4 ROTATED 180 DEGREES
======================================================================
Original obstacle count: 57
Rotated obstacle count: 57
START: (0, 0)
GOAL: (19, 19)

Environment verification:
Grid: 20 x 20
Obstacles: 57
Shortest valid path: 38
Reachable states: 343

================================================================================
EXPERIMENT 6 TRAINING STARTED
================================================================================

Training: Sparse Q-Learning
 Seed 1/10 completed in 14.0 sec
 Seed 2/10 completed in 8.2 sec
 Seed 3/10 completed in 5.8 sec
 Seed 4/10 completed in 5.8 sec
 Seed 5/10 completed in 7.9 sec
 Seed 6/10 completed in 7.9 sec
 Seed 7/10 completed in 6.5 sec
 Seed 8/10 completed in 7.9 sec
 Seed 9/10 completed in 10.3 sec
 Seed 10/10 completed in 6.9 sec
Sparse Q-Learning completed in 81.2 seconds

Training: Fixed PBRS Manhattan
 Seed 1/10 completed in 59.2 sec
 Seed 2/10 completed in 60.0 sec
 Seed 3/10 completed in 5.0 sec
 Seed 4/10 completed in 6.7 sec
 Seed 5/10 completed in 57.8 sec
 Seed 6/10 completed in 5.9 sec
 Seed 7/10 completed in 8.3 sec
 Seed 8/10 completed in 6.4 sec
 Seed 9/10 completed in 58.5 sec
 Seed 10/10 completed in 58.0 sec
Fixed PBRS Manhattan completed in 325.8 seconds

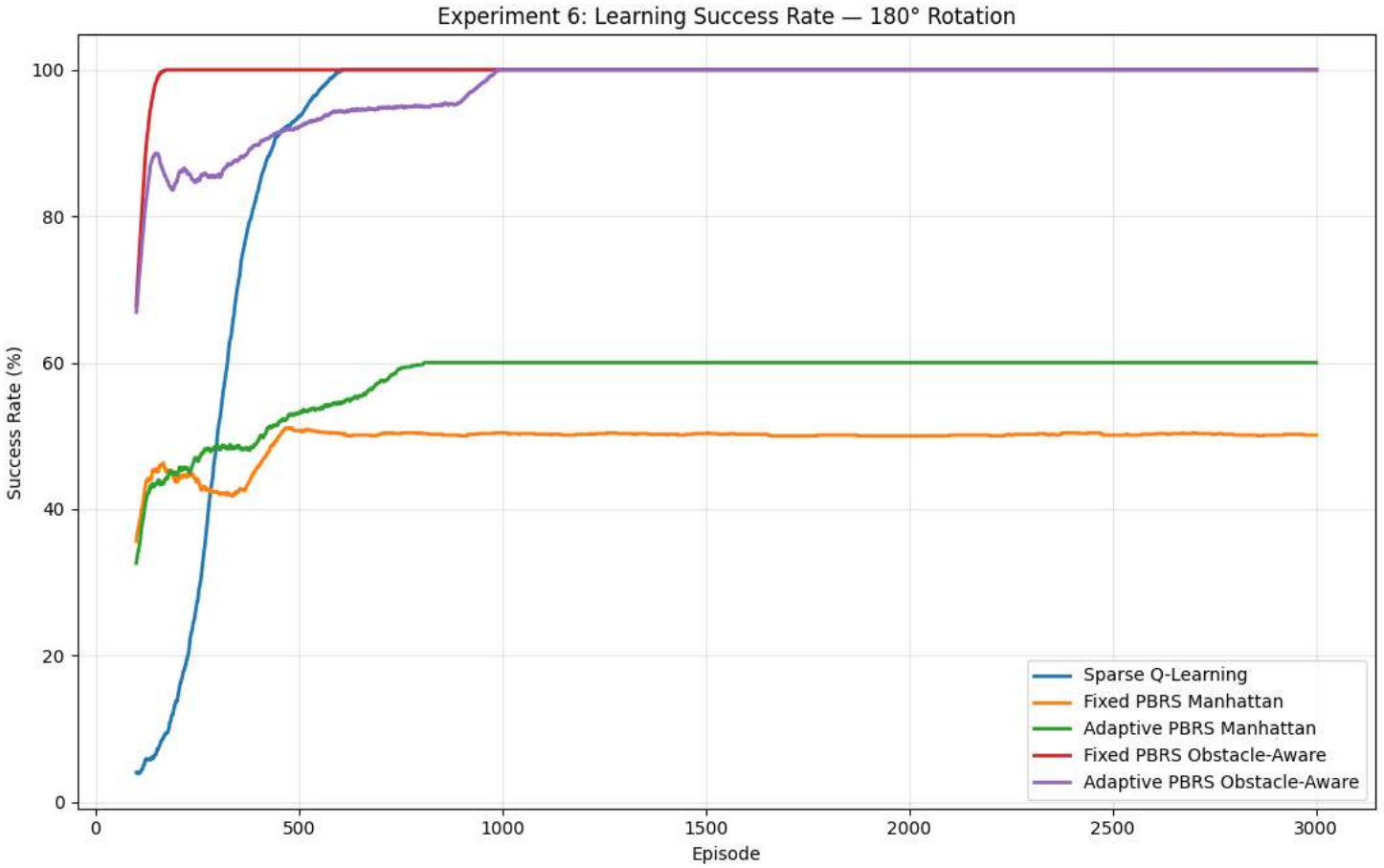
Training: Adaptive PBRS Manhattan
 Seed 1/10 completed in 58.9 sec
 Seed 2/10 completed in 7.1 sec
 Seed 3/10 completed in 6.6 sec
 Seed 4/10 completed in 5.4 sec
 Seed 5/10 completed in 58.9 sec
 Seed 6/10 completed in 5.4 sec
 Seed 7/10 completed in 12.3 sec
 Seed 8/10 completed in 5.8 sec
 Seed 9/10 completed in 58.9 sec
 Seed 10/10 completed in 58.3 sec
Adaptive PBRS Manhattan completed in 277.5 seconds

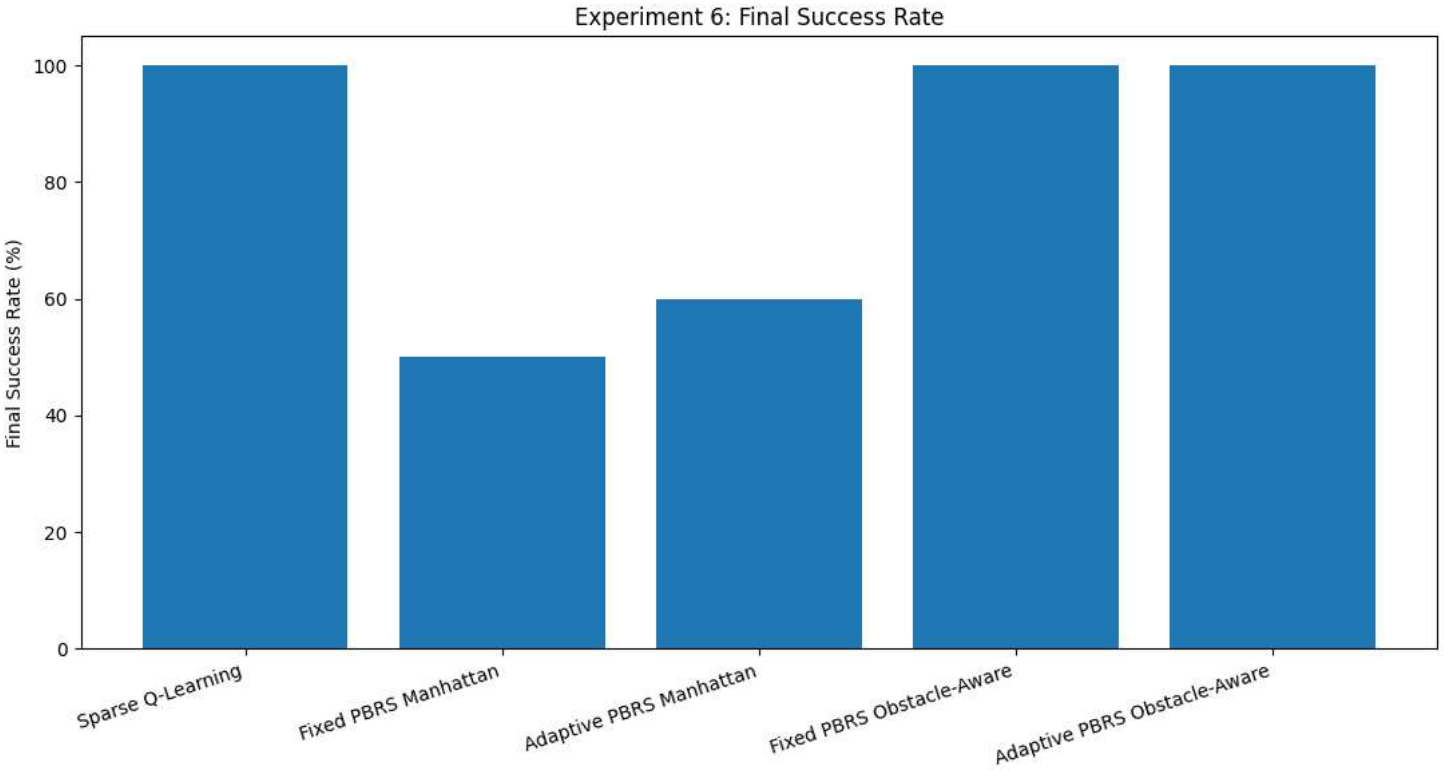
Training: Fixed PBRS Obstacle-Aware
 Seed 1/10 completed in 5.6 sec
 Seed 2/10 completed in 4.4 sec
 Seed 3/10 completed in 4.4 sec
 Seed 4/10 completed in 5.8 sec
 Seed 5/10 completed in 4.3 sec
 Seed 6/10 completed in 4.8 sec
 Seed 7/10 completed in 5.2 sec
 Seed 8/10 completed in 4.3 sec
 Seed 9/10 completed in 5.3 sec
 Seed 10/10 completed in 4.6 sec
Fixed PBRS Obstacle-Aware completed in 48.9 seconds

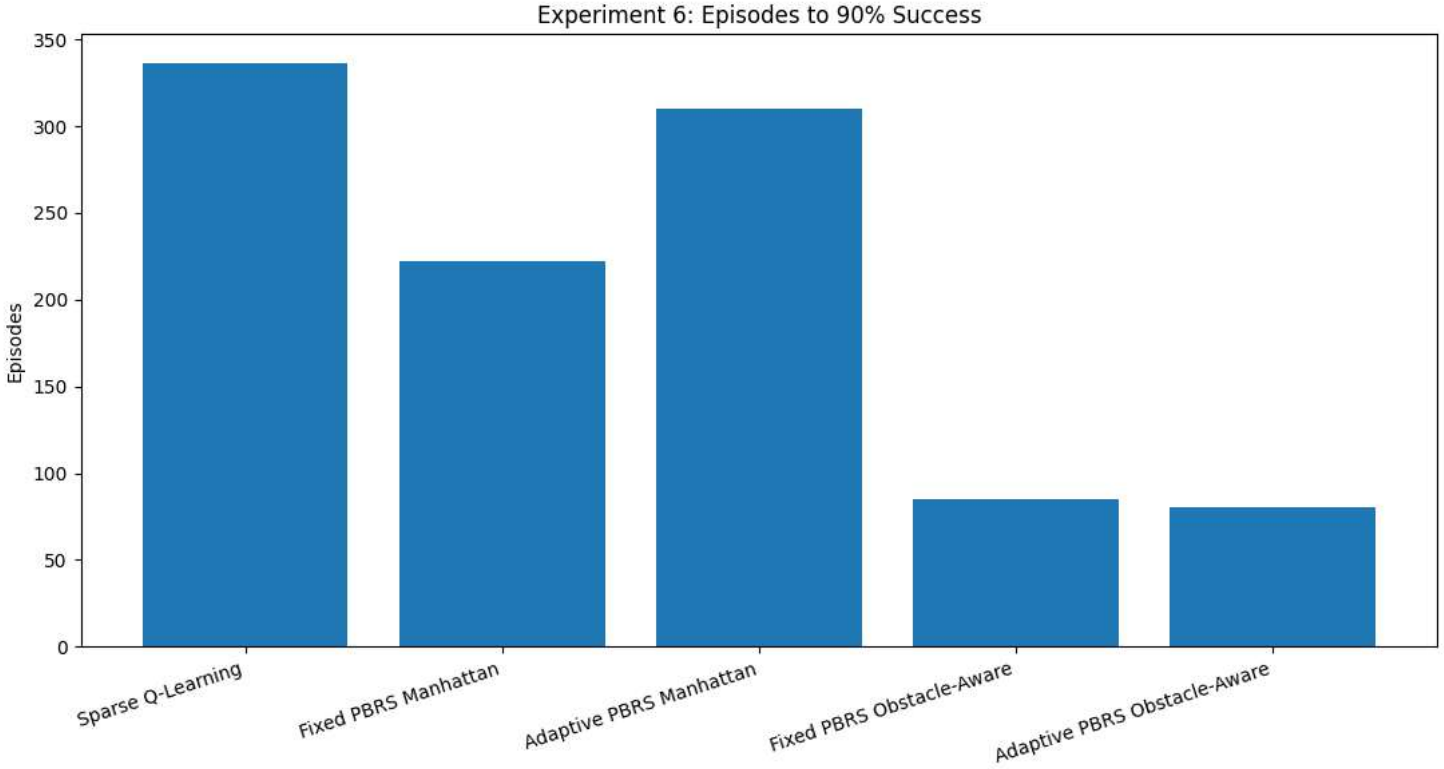
Training: Adaptive PBRS Obstacle-Aware
 Seed 1/10 completed in 15.1 sec
 Seed 2/10 completed in 6.0 sec
 Seed 3/10 completed in 5.6 sec
 Seed 4/10 completed in 6.4 sec
 Seed 5/10 completed in 6.6 sec
 Seed 6/10 completed in 5.1 sec
 Seed 7/10 completed in 6.5 sec
 Seed 8/10 completed in 5.2 sec
 Seed 9/10 completed in 6.2 sec
 Seed 10/10 completed in 5.6 sec
Adaptive PBRS Obstacle-Aware completed in 68.4 seconds

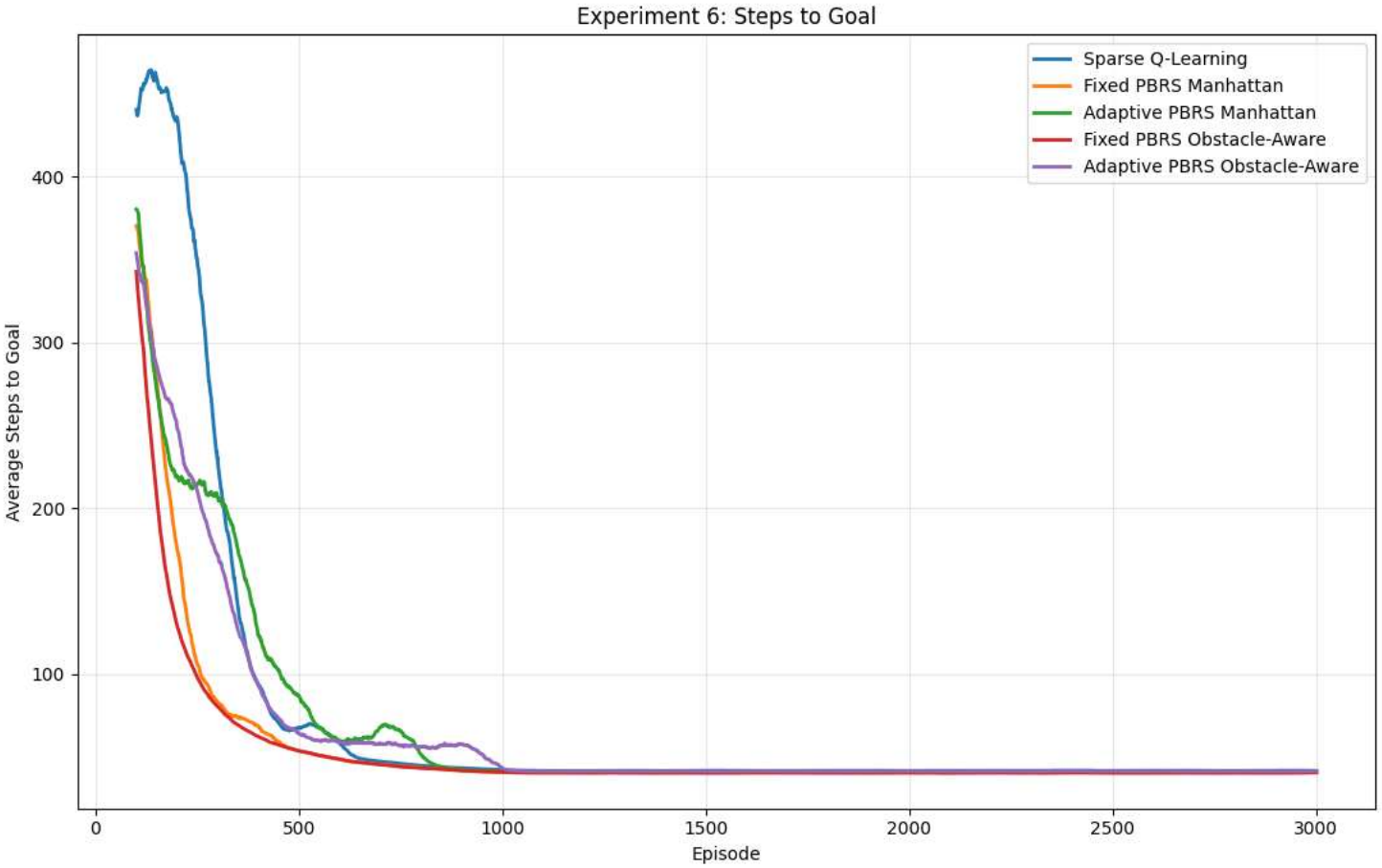
================================================================================================
EXPERIMENT 6 — FINAL RESULTS
================================================================================================
```

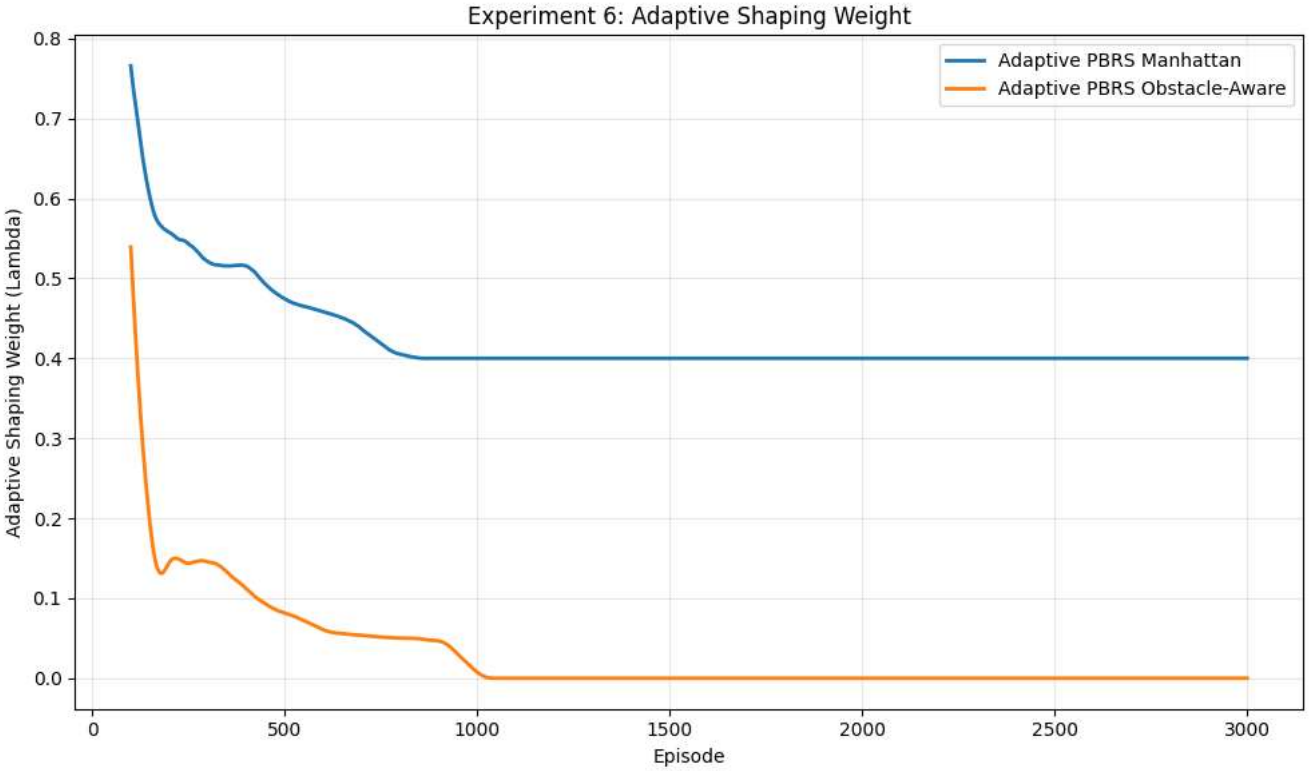
	Method	Final Success Rate (%)	Success SD	Average Steps to Goal	First Successful Episode	Seeds Reaching 90%	Episodes to 90% Success
0	Sparse Q-Learning	100.0	0.00	41.46	23.1	10	336.20
1	Fixed PBRS Manhattan	50.1	52.60	40.02	15.7	5	222.20
2	Adaptive PBRS Manhattan	60.0	51.64	40.69	15.7	6	309.83
3	Fixed PBRS Obstacle-Aware	100.0	0.00	40.24	14.4	10	84.70
4	Adaptive PBRS Obstacle-Aware	100.0	0.00	41.11	14.4	10	80.80

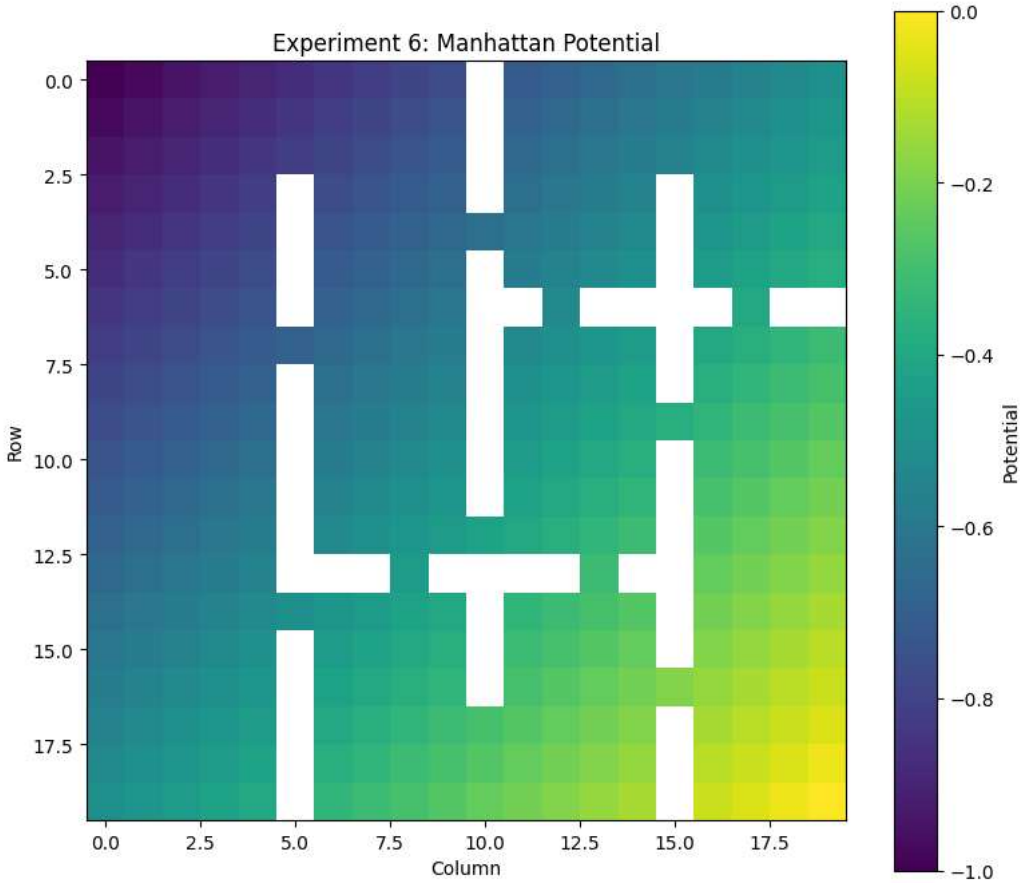


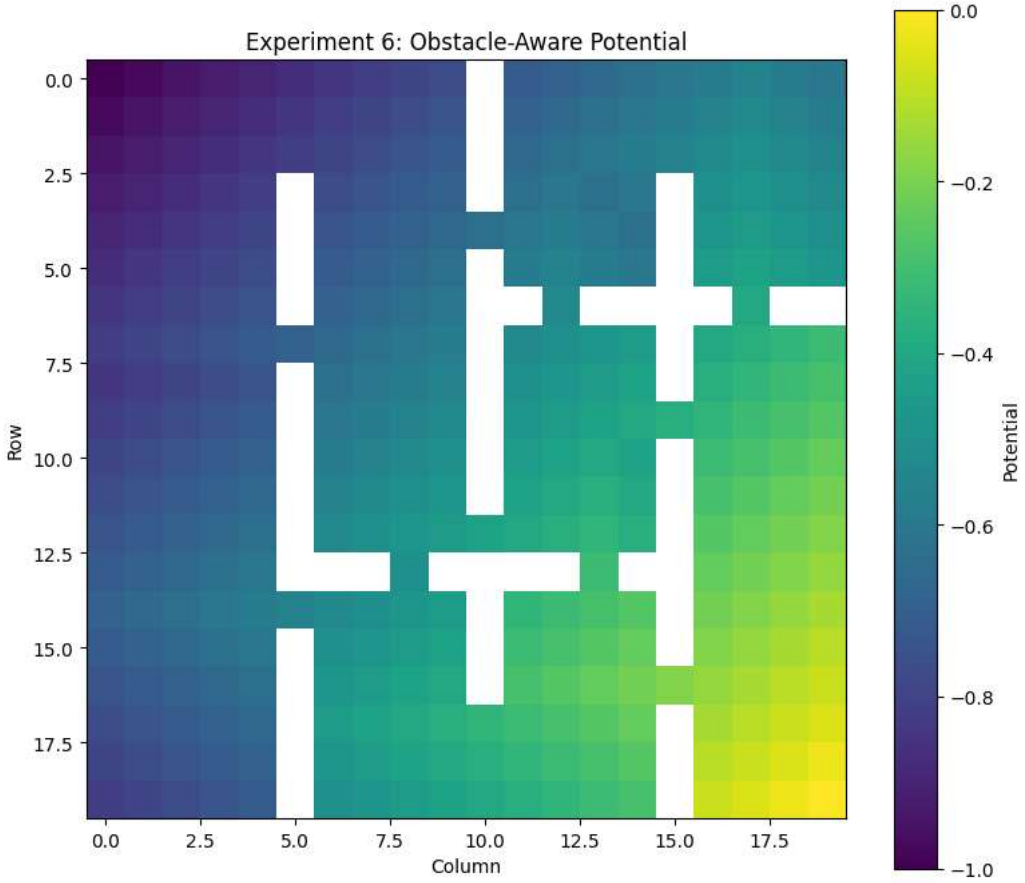

Experiment 6: Final Success Rate
Final Success Rate (%)
0
20
40
60
80
100
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware

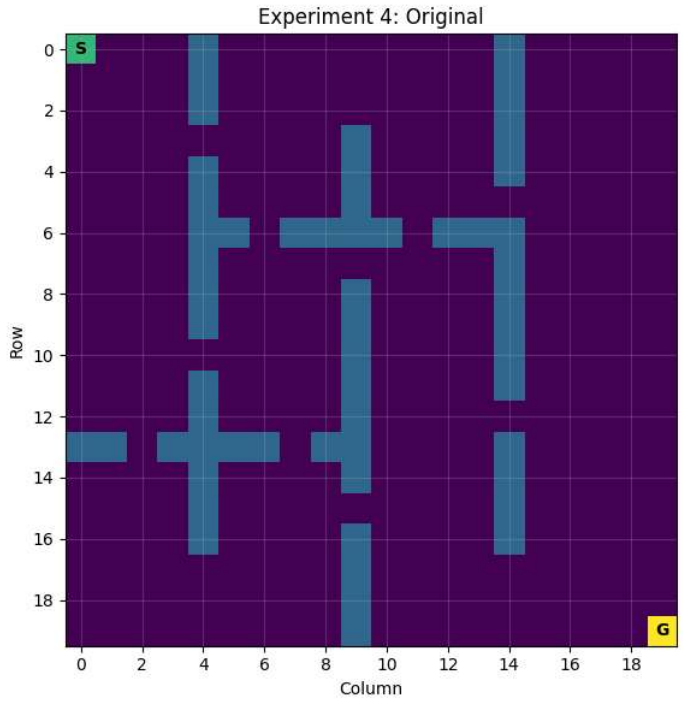

Experiment 6: Episodes to 90% Success
Episodes
0
50
100
150
200
250
300
350
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware

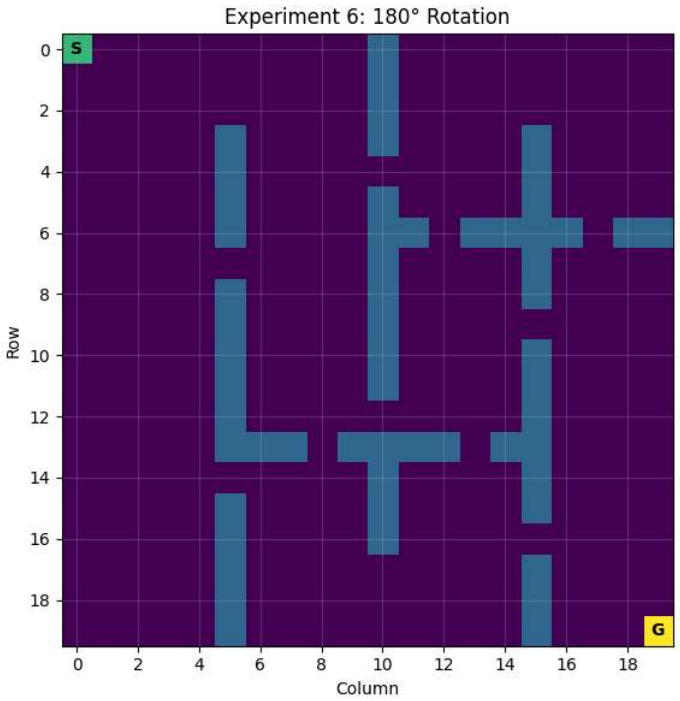

Experiment 6: Steps to Goal
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware
Average Steps to Goal
Episode
100
200
300
400
0
500
1000
1500
2000
2500
3000


Experiment 6: Adaptive Shaping Weight
Adaptive PBRS Manhattan
Adaptive PBRS Obstacle-Aware
Adaptive Shaping Weight (Lambda)
Episode
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
500
1000
1500
2000
2500
3000


Experiment 6: Manhattan Potential
Row
Column
Potential
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−0.2
−0.4
−0.6
−0.8
−1.0


Experiment 6: Obstacle-Aware Potential
Row
Column
Potential
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−0.2
−0.4
−0.6
−0.8
−1.0






```
================================================================================================
EXPERIMENT 6 COMPLETED
================================================================================================
Rotation: 180 degrees from ORIGINAL Experiment 4
Training episodes: 3000
Seeds: 10
Shortest path: 38
Total execution time: 13.51 minutes

FINAL TABLE:

                      Method  Final Success Rate (%)  Success SD  Average Steps to Goal  First Successful Episode  Seeds Reaching 90%  Episodes
          Sparse Q-Learning                   100.0        0.00                  41.46                      23.1                  10
       Fixed PBRS Manhattan                    50.1       52.60                  40.02                      15.7                   5
    Adaptive PBRS Manhattan                    60.0       51.64                  40.69                      15.7                   6
   Fixed PBRS Obstacle-Aware                   100.0        0.00                  40.24                      14.4                  10
Adaptive PBRS Obstacle-Aware                   100.0        0.00                  41.11                      14.4                  10

Generated files:
1. experiment6_summary.csv
2. experiment6_seed_results.csv
3. exp6_success_rate.png
4. exp6_final_success.png
5. exp6_convergence.png
6. exp6_steps.png
7. exp6_lambda.png
8. exp6_manhattan_potential.png
9. exp6_obstacle_potential.png
10. exp6_environment_comparison.png
```